source ..... noise engineering manual
rev 2026.10

virt iter legio

_travelers guide
_for oscillating in stereo

3 algorithms	2 pm inputs	6 hp

[enc] press [enc] hold › then noon = 12 o'clock tags = the panel drawing

three-algorithm stereo oscillator · stereo pm ins · chorus · the legio launch firmware

← back to the legio starsystem

orange = hold

01 what it is — vil_01

an oscillator	three stereo algorithms · harm, sawx, bass · plus a vintage-style chorus
family	the same algorithms live in the arturia microfreak and the virt vereor plugin
out L R	stereo out, up to 16 V peak to peak
platform	the legio launch firmware · other firmwares

02 pitch — vil_02 · enc

enc [pitch]	turn = fine tune
[enc] + turn	push and turn = coarse tune
enc jack	1 V/oct pitch, –2 V to +5 V
tuning	auto-calibrates at power-on · leave the pitch jack empty while it boots

10 first flight — vil_10 · a clean start

1. power up with enc jack empty so it calibrates
2. out L + R to your mixer · s2 0 · s1 harm
3. 1 [flavor] and 2 [tang] fully left
4. tune with enc [pitch] · [enc] + turn for big jumps
5. sweep flavor, then tang · stop before the last sixth of tang
6. try sawx and bass · then add chorus with s2

__fig.04 : the algorithm map — one switch, two knobs

s1	1 flavor	2 tang
harm (up)	partial spacing	rectify partials
sawx (mid)	modulus gain	stereo phase
bass (down)	cos saturation	two-stage fold

03 harm — vil_03 · s1 up · fig.01

what	additive sines with a touch of digital grit
1 [flavor]	partial spacing · 0 = unison · max = octaves · between glides in frequency
2 [tang]	rectifies each partial, like half a wavefolder

__fig.01 : harm flavor — where the partials sit



04 sawx — vil_04 · s1 mid

what	a supersaw with "saw-mod" · ethereal to metallic
1 [flavor]	gain into a modulus stage
2 [tang]	how much stereo phase

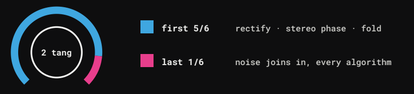
05 bass — vil_05 · s1 down

what	quadrature modulation after electronotes #73, plus ne extras
1 [flavor]	saturation of the cos oscillator
2 [tang]	two-stage asymmetric wavefolder

06 tang · top sixth — vil_06 · fig.02

every algorithm	the last sixth of tang adds noise
harm	phase-modulated noise mixed in
sawx	pm from subsampled white noise
bass	noise between the fold stages, also pm on both oscillators

__fig.02 : tang — clean, then noisy



07 switches · jacks — vil_07 · s1 · s2

s1	[harm] up · [sawx] mid · [bass] down
s2	[II] lots of chorus · [I] some · [0] off
trig sync	hard sync on a rising edge near +1.6 V
in pm L R	stereo phase mod · ac-coupled audio · L feeds R if alone
1 2 jacks	cv for flavor and tang, 0 – 5 V

08 four patches — vil_08 · from the manual

drone — any
out L + R straight to the mixer · move flavor and tang by hand or cv

voice — any
outs into two vcas · an envelope on both · pitch cv into enc · envelopes or lfos on flavor and tang

phase mod — any
mult the pitch cv to a second oscillator · its output through an attenuator into pm L

mono — any
just out L into one vca

__fig.05 : signal path — as the manual describes it



__fig.03 : the panel — numbers match the legend



enc pitch	turn fine · push + turn coarse
1 flavor	per algorithm · fig.04
2 tang	per algorithm · top 1/6 noise
s1 algorithm	harm · sawx · bass
s2 chorus	II · I · 0
trig sync	hard sync · ~1.6 V
cv pitch · 1 · 2	pitch –2…+5 V · others …
in pm L / R	phase mod · L feeds R
out out L / R	stereo out

09 specs — vil_09

pitch cv	1 V/oct · –2 V to +5 V
cv in	0 – 5 V
sync	rising edge near +1.6 V
size	6 hp · 0.9 in deep
power	140 mA +12 V · 22 mA –12 V · no 5 V

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · panel drawing after the noise engineering legio overlay template
not affiliated with noise engineering, mathias kettner or marvel

conceived by mellotom · legio-virt-iter-navigator ... 00.10_01
created with claude